THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a perfect square? 1

(a) 200 **(b)** 144 **(c)** 99 **(d)** 150

2. The value of 13^2 is: 1

(a) 169 **(b)** 182 **(c)** 156 **(d)** 26

3. $\sqrt{196}$ is equal to: 1

(a) 13 **(b)** 14 **(c)** 15 **(d)** 16

4. $\sqrt{1089}$ is equal to: 1

(a) 33 **(b)** 32 **(c)** 35 **(d)** 34

5. The sum of the first 15 odd natural numbers is: 1

(a) 30 **(b)** 224 **(c)** 225 **(d)** 240

6. How many non-square numbers lie between 14^2 and 15^2? 1

(a) 28 **(b)** 14 **(c)** 29 **(d)** 27

7. The value of 15^3 is: 1

(a) 3390 **(b)** 3360 **(c)** 675 **(d)** 3375

8. $\sqrt[3]{3375}$ is: 1

(a) 16 **(b)** 15 **(c)** 17 **(d)** 14

9. Which of the following is a perfect cube? 1

(a) 900 **(b)** 1000 **(c)** 1200 **(d)** 800

10. The ones digit of 27^2 is: 1

(a) 1 **(b)** 4 **(c)** 7 **(d)** 9

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Is 576 a perfect square? Give its square root. 2

12. Is 1225 a perfect square? Give its square root. 2

13. Is 2304 a perfect square? Give its square root. 2

14. Write the sum of the first 5 odd natural numbers without adding. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the cube root of 3375 by prime factorisation. 3

16. Find the cube root of 5832 by prime factorisation. 3

17. Find the cube root of 10648 by prime factorisation. 3

OR

Find the cube root of 17576 by prime factorisation. 3

18. Find $\sqrt{2025}$ using prime factorisation. 3

OR

Find $\sqrt{3136}$ using prime factorisation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Show that $1 + 3 + 5 + 7 + 9 + 11 = 6^2$. 5
(ii) Find the sum of the first 25 odd numbers.
(iii) Which number's square is the sum of the first 40 odd numbers?

OR

(i) Find $\sqrt{7569}$ by prime factorisation. 5
(ii) Find the least number to multiply 7569 to get a perfect cube.

20. (i) Find the cube root of 12167. 5
(ii) Find the smallest number to divide 3240 to make it a perfect cube.

OR

Case study: A school hall is a square with side 24 m and is to be tiled with square tiles of side 0.5 m. 5
(i) Find the area of the hall.
(ii) How many tiles are needed?
(iii) If a cube-shaped water tank has edge 1.5 m, find its volume in litres.

END OF QUESTION PAPER — Total: 40 Marks

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